

A

Preparation, then queries

task snapshot (no model call)

issue text, repository tree at the pinned commit,
reproducer command, ranked source-file budget

static extraction (parsers only)

Tree-sitter for Python, Joern code property graphs for C/C++/Fortran/Cython; import and dynamic-binding resolution; findings, conditions and documented definitions captured with source spans. Nothing is executed here.

queryable evidence store

computations: name, path, result anchor, expressions
conditions and findings; documented public definitions
call boundaries: internal | external provider | unknown

agent endpoints (same budget as baseline)

science_find - Find scientific computations or public evidence by name, phrase or path.
 science_inspect - relationships, definitions, source, contracts
 science_note - optional record: never gates repair

B

Payloads the agent sees

one recorded boundary contract: task 103, shell.py:1005

```

{"callee": "radians", "code": "np.radians(thetad)",
 "boundary": {"kind": "external", "provider": "numpy",
              "assume": "correct_interface",
              "repair_scope": "caller_file"}}

```

science_find response shape (keys verbatim)

```
{
  "status": "ok",
  "matches": [{"target": "...",
    "excerpt": "...",
    "kind": "scientific_node"}],
  "total_matches": N,
  "next_offset": ...,
  "scope": "Lexical discovery, not a verdict."}
```

the same call in the baseline arm

```
grep -rn "radians" .      # provider, contract and repair
                           # scope are rebuilt by reading the
                           # file and its imports; nothing is
                           # marked assumed-correct
```